5	(a)	Define *potential difference between two points in a circuit.*	
		....	
		....	[1]
		Solution: It is the amount of electrical energy converted to other forms of energy per unit charge flowing between those two points in a circuit.	1
	(b)	A thermistor and a variable resistor are connected in a potential divider circuit as shown in Fig. 5.1. The battery has an e.m.f. of E, the thermistor has a resistance R_T and the variable resistor has a resistance R_V. This circuit is used to activate an alarm system whenever the ambient temperature rises to a certain value. The alarm bell will sound if the potential difference across it increases beyond the pre-set value.	

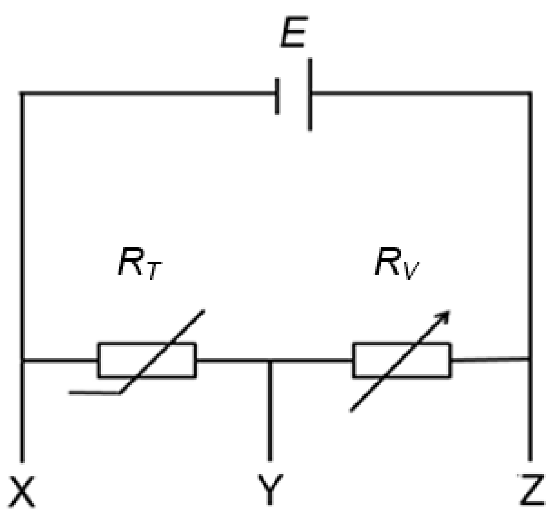

Fig. 5.1

State an expression for the potential difference across the variable resistor in terms of E, R_T and R_V.

[1]

Solution:

Let the current in the circuit be I

e.m.f. of cell $E = I (R_T + R) \Rightarrow I = E / (R_T + R_V)$

p.d. across variable resistor $= I R_V = E [R_V / (R_T + R_V)]$ 1

OR

By potential divider principle,

p.d. across variable resistor $= E [R_V / (R_T + R_V)]$

(i) State and explain across which two terminals, XY or YZ, how the alarm bell should be connected to the circuit in Fig. 5.1.

………………………………………………………………………………………………………

………………………………………………………………………………………………………

………………………………………………………………………………………………………

……………………………………………………………………………………………… [2]

Solution:

As the temperature increases, the resistance of the thermistor R_T decreases As R_T decreases, the potential difference across the variable resistor increases since p.d. across variable resistor $= I R_V = E R_V / (R_T + R_V)$ where E and R_v are constants. 1

Hence if the alarm bell is to be triggered when the p.d. across it goes beyond a particular value, the alarm should be connected across YZ (variable resistor). 1

(ii) State and explain the purpose of the variable resistor in the circuit.

………………………………………………………………………………………………………

………………………………………………………………………………………………………

………………………………………………………………………………………………………

……………………………………………………………………………………………… [2]

Solution:

The variable resistor controls the temperature at which the alarm to be triggered. 1

The p.d. across the resistor depends on the resistance of the variable as well as that of the thermistor. By changing its resistance, it allows the p.d. across it to reach the triggering value at other temperatures when the thermistor reaches a different temperature. 1

(c) Fig. 5.2 shows a potentiometer circuit consisting of a 100 cm length of wire AB and a driver cell E_1 of e.m.f 2.0 V and of negligible internal resistance.

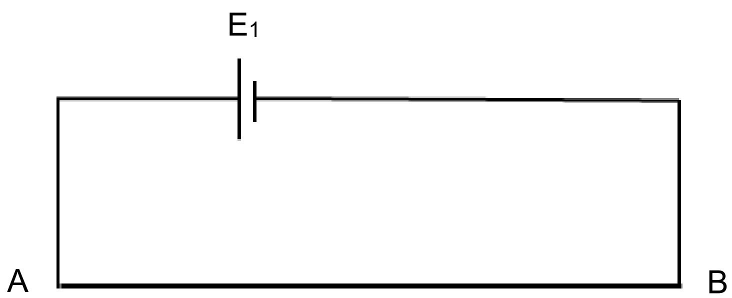

Fig. 5.2

(i) State the value of the potential gradient along AB.

potential gradient = ………….. V cm^{-1} [1]

Solution:

Potential gradient = 2.0 / 100 = 0.020 V cm^{-1} 1

(ii) A circuit consisting of a cell E_2 and a variable resistor R is now connected to the potentiometer as shown in Fig. 5.3.

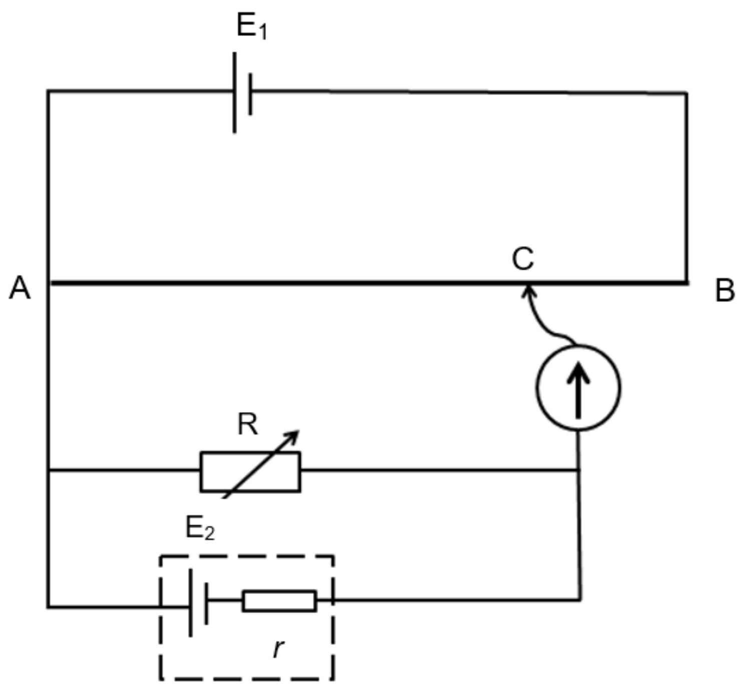

Fig. 5.3

	(d)	The resistance of R is set to 10.0 Ω and the sliding contact C is adjusted until there is no current flowing in the galvanometer. The length AC is found to be 60.0 cm. The resistance of R is now set to 3.0 Ω and the experiment is repeated. The length AC is now 54.0 cm.	
		Calculate the e.m.f. and internal resistance *r* of cell E_2.	
		e.m.f. of cell E_2 = ……………………..V internal resistance of cell E_2 = …………………. Ω	[5]
		Solution: p.d. across AB = e.m.f. of E_1 = 2.0 V When resistance of R = 10.0, balance length = 60.0 cm p.d. across AC = potential gradient x 60.0 = 0.020 x 60.0 = 1.2 V p.d. across R = E_2 x 10.0 / (10.0 + r) At balance length, p.d. across R = p.d. across AC E_2 x 10.0 / (10.0 + r) = 1.2 ………..(1) When resistance of R = 3.0, balance length = 54.0 cm p.d. across AC = potential gradient x 54.0 = 0.020 x 54.0 = 1.08 V p.d. across R = E_2 x 3.0 / (3.0 + r) At balance length, p.d. across R = p.d. across AC E_2 x 3.0 / (3.0 + r) = 1.08 ………..(2) (1) / (2) (10.0 /3.0) x [(3.0 + r) / (10.0 + r)] = 1.2 / 1.08 r = 0.5 Ω subtitute r = into (1) E_2 = 1.26 V	 1 1 1 1 1

6	**(a)**	A coil with 500 turns is placed in a uniform magnetic field of flux density 5.0×10^{-2} T. The area of